

ASSIGNMENT-20

HALLTICKET: 2303A51118

BATCH:03

Lab 20 : Security Testing : Identifying Vulnerabilities in AI – Generated Code.

Task – 01 : Password Security Enhancement.

Prompt : Generate a Python user registration and login script that stores usernames and passwords.

Code & Output :



The screenshot shows a code editor interface with a file explorer on the left and a terminal at the bottom. The main editor area displays a Python script titled 'TASK 01'. The script uses the 'bcrypt' library for password hashing and verification. It defines a 'register' function that hashes a password and stores it in a dictionary, and a 'login' function that checks the username and password against the stored hashes. The script then demonstrates the registration of an 'admin' user and two login attempts: one successful and one with incorrect credentials.

```
import bcrypt

users = {}

def register(username, password):
    hashed = bcrypt.hashpw(password.encode(), bcrypt.gensalt())
    users[username] = hashed

def login(username, password):
    if username in users and bcrypt.checkpw(password.encode(), users[username]):
        return "Login Successful"
    return "Invalid Credentials"

register("admin", "password123")
print(login("admin", "password123"))
print(login("admin", "wrong"))
```

*** Login Successful
Invalid Credentials

Explanation :

Plain text passwords are unsafe because they can be easily exposed in data breaches. Hashing converts passwords into irreversible values, improving security. bcrypt adds salting, preventing attacks like rainbow tables. The updated script securely verifies login without storing real passwords.

Task – 02 : Secure API Key Management.

Prompt : Generate a Python script that fetches data from an API using an API key.

Code :



```
TASK 02

import os
import requests

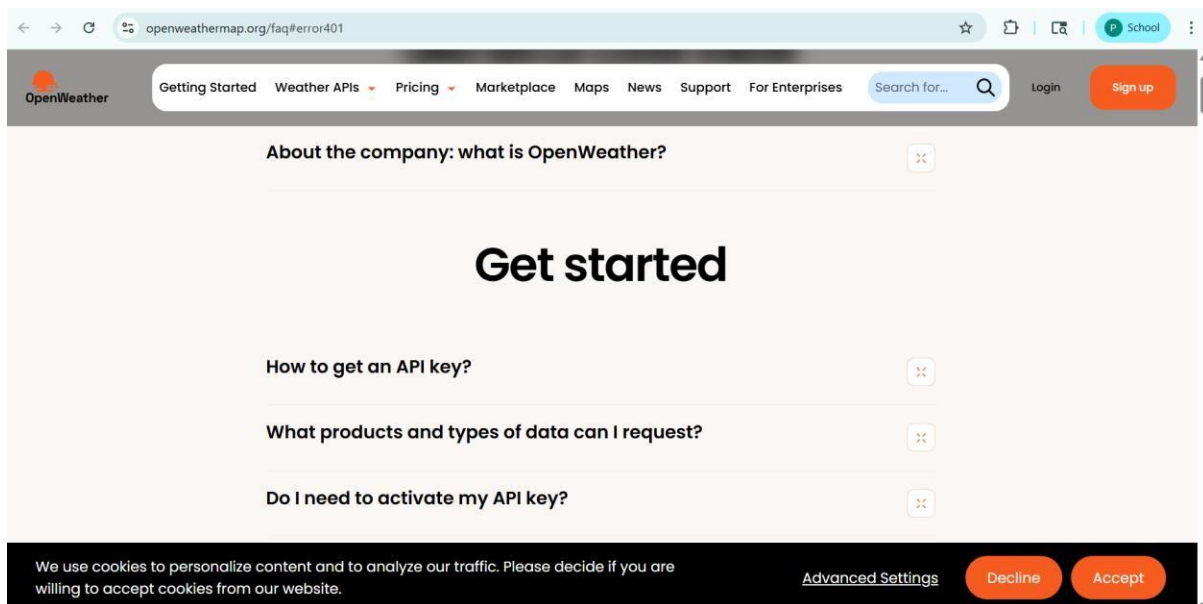
api_key = os.getenv("API_KEY")

url = f"https://api.openweathermap.org/data/2.5/weather?q=London&appid={api_key}"
response = requests.get(url)

print(response.json())

... {'cod': 401, 'message': 'Invalid API key. Please see https://openweathermap.org/faq#error401 for more info.'}
```

Output :



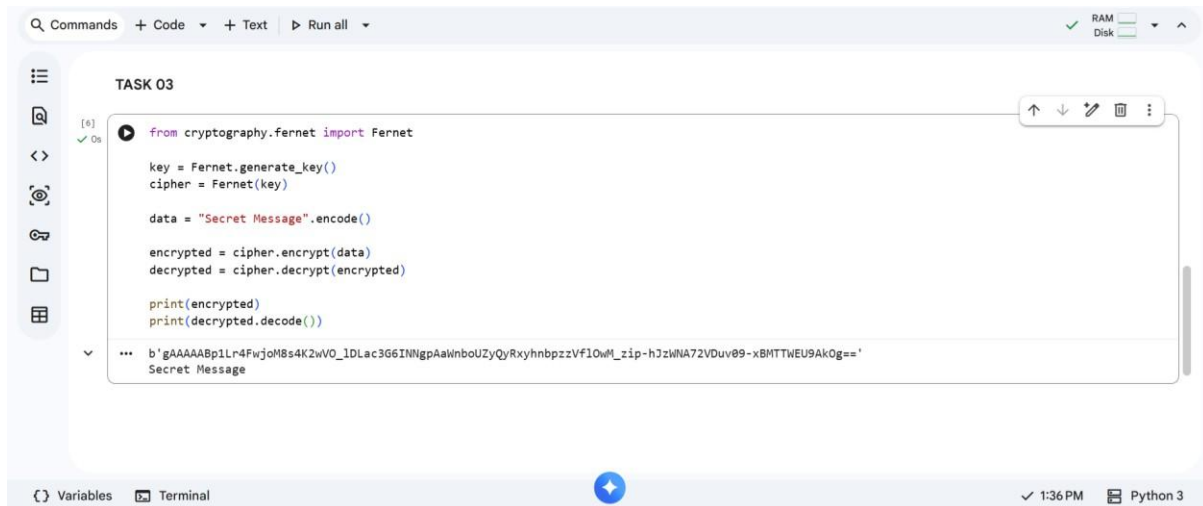
Explanation :

Hardcoding API keys exposes them in source code and version control. Environment variables securely store sensitive data outside the code. This prevents unauthorized access and improves deployment security. The script still works normally while keeping credentials hidden.

Task – 03 : Weak Encryption Detection.

Prompt : Generate a Python encryption-decryption program using a simple method.

Code & Output :



The screenshot shows a code editor interface with a file named 'TASK 03'. The code is as follows:

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()
cipher = Fernet(key)

data = "Secret Message".encode()

encrypted = cipher.encrypt(data)
decrypted = cipher.decrypt(encrypted)

print(encrypted)
print(decrypted.decode())
```

The output of the program is displayed below the code:

```
b'gAAAAABp1Lr4Fwj0M8s4K2wV0_IDLac366INNgpAaWnboUZyQyRxyhnbpzVf10wM_zip-hJzWNA72VDuv09-xBMTTWEU9AkOg=='
Secret Message
```

The interface also shows a 'Variables' panel at the bottom left and a 'Terminal' panel at the bottom right, which is currently empty. The status bar at the bottom indicates the time is 1:36 PM and the environment is Python 3.

Explanation :

Basic encoding like base64 is reversible and not secure encryption. Fernet provides strong symmetric encryption using secure algorithms. Data cannot be decrypted without the correct key. This ensures confidentiality and protects sensitive information.

Task – 04 : Real Time Application : Security Review of AI – Generated Web Service.

Prompt : Generate a simple Flask API for user login.

Code & Output :



```
[9] from flask import Flask, request, jsonify
import bcrypt

app = Flask(__name__)
users = {
    "admin": bcrypt.hashpw("password123".encode(), bcrypt.gensalt())
}
@app.route('/login', methods=['POST'])
def login():
    data = request.json
    username = data.get("username")
    password = data.get("password")
    if username in users and bcrypt.checkpw(password.encode(), users[username]):
        return jsonify({"message": "Success"})
    return jsonify({"message": "Unauthorized"}), 401

if __name__ == '__main__':
    app.run()

... * Serving Flask app '__main__'
* Debug mode: off
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
INFO:werkzeug:Press CTRL+C to quit
```

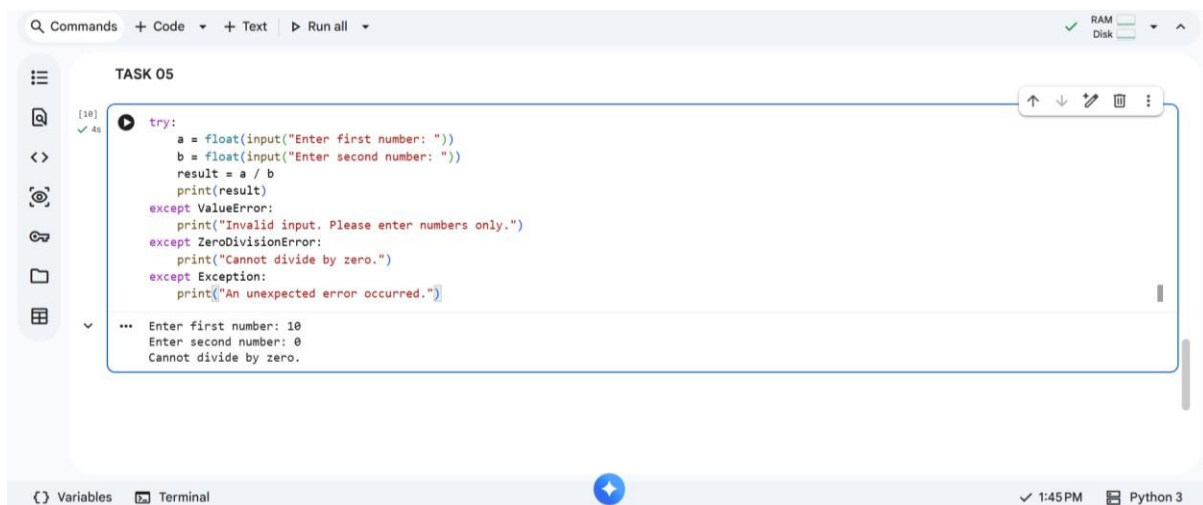
Explanation :

Common vulnerabilities include plain-text passwords, no validation, and insecure responses. bcrypt ensures passwords are securely stored and verified. Input is handled safely using JSON parsing. The API now reduces authentication risks and improves overall security.

Task – 05 : Exception Handling Improvement.

Prompt : Generate a Python program that takes two numbers and performs division.

Code & Output :



```
TASK 05

[10] try:
✓ 4s a = float(input("Enter first number: "))
      b = float(input("Enter second number: "))
      result = a / b
      print(result)
except ValueError:
    print("Invalid input. Please enter numbers only.")
except ZeroDivisionError:
    print("Cannot divide by zero.")
except Exception:
    print("An unexpected error occurred.")

... Enter first number: 10
Enter second number: 0
Cannot divide by zero.
```

Explanation :

Programs without exception handling may crash during runtime errors. Invalid inputs and division by zero are common issues. try-except blocks ensure smooth execution and prevent crashes. Users receive clear error messages instead of program failure.